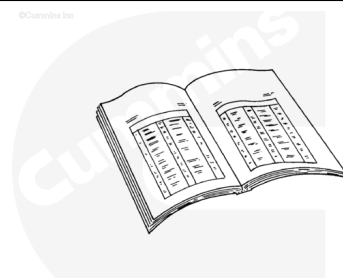


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>)
- Remove the oil suction tube and transfer tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html>)
- Remove the oil pump, if necessary. Refer to Procedure 007-031 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)

- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.

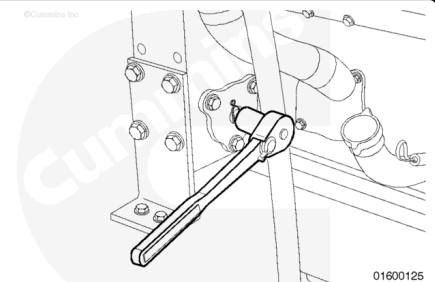
(/qs3/pubsys2/xml/en/procedures/56/56-001-046-tr.html)

Note : Use the following procedure for more information on connecting rod types. Refer to Procedure 001-014 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-014.html)

Remove

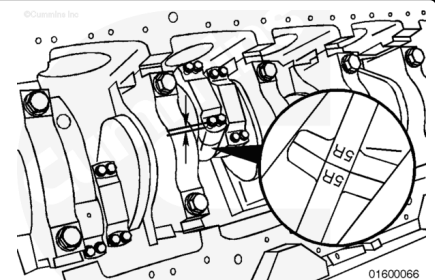
Use the barring mechanism to rotate the engine crankshaft to position a connecting rod at bottom dead center for maximum accessibility when loosening the connecting rod capscrews.

Bar the engine to the desired position.



The connecting rods **must** have the cylinder number and bank marked on both the connecting rod and the connecting rod cap on the inboard side. Check the connecting rods for the correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

Loosen the capscrews until there is 6 mm [0.25 in] of clearance between the rod cap and the capscrew head.



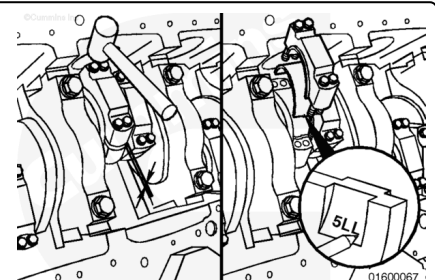
Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate.

Remove the capscrews and the cap.

Mark the cylinder number, right (R) or left (L) bank, and the letter "L" in the flat surface of the bearing tangs.

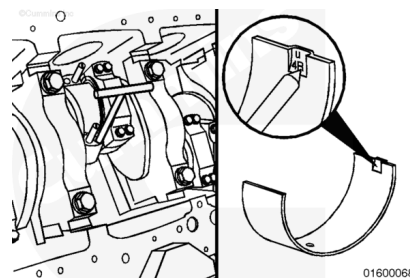
Use an awl to mark the bearing position in the tang area.

Remove the lower rod bearing.



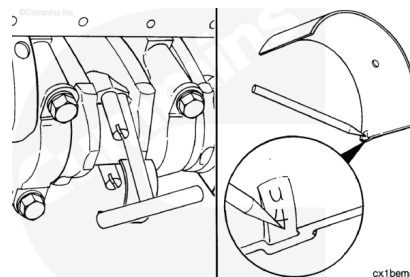
Push the connecting rod up far enough to allow the upper bearing shell to be removed. Push the rod away from the crankshaft.

Remove the bearing shell, and mark the cylinder number, right (R) or left (L) bank, and the letter "U" on the flat surface of the bearing tangs.



The bearings **must** be marked for location as they are removed for future identification. Each bearing **must** be installed in its original location if the bearing is installed again.

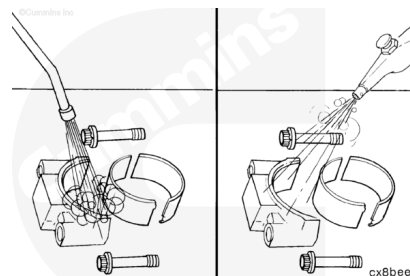
Use a scribe or ball point pen to mark the bearing location in the tang area.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

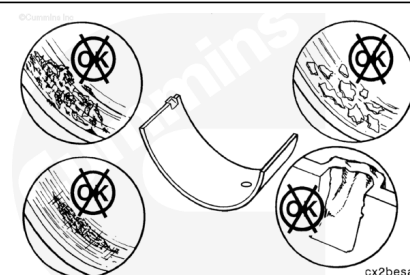
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the bearings.

Dry with compressed air.

Inspect the bearing face on the front and back for:

- Nicks
- Cracks
- Burrs
- Scratches (deep enough to be felt with a fingernail)
- Scrapes



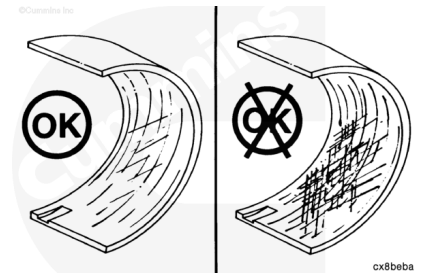
- Gouges
- Embedded particles
- Fretting
- Pitting
- Flaking
- Corrosion
- Lock tang damage.

If any damage is found, the bearing **must** be replaced.

Normal bearing wear produces a smooth finish which will wear into the copper lining. Exposed copper does **not always** indicate worn bearings. Reference the Parts Reuse Guidelines, Bulletin 3810303.

If large areas of copper lining are visible in the bearings before the engine has accumulated 3000 hours, inspect the engine for contamination from fine dirt particles and correct the problem.

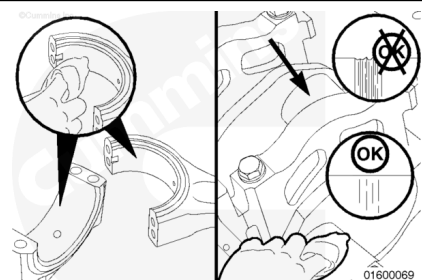
For more detailed information of bearing damage, reference the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387



Install

Use a clean, lint-free paper towel or cloth to clean the rod bore, the cap, and the crankshaft rod journal.

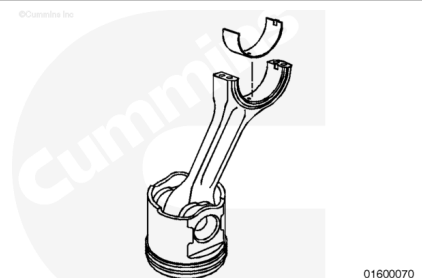
Inspect the crankshaft rod journals for damage.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the back of the bearing shells.

Note : The illustration shows the piston and connecting rod removed from the block for clarity.



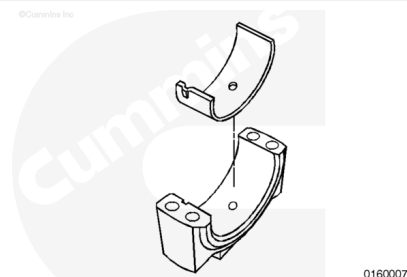
Used bearings **must** be installed in the same location from which they were removed.

Install the upper bearing shell in the connecting rod with the tang of the bearing in the slot of the rod.

Lubricate the bearing shell with Lubriplate™ 105, Part Number 3163086.

Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the rod cap, and the end of the bearing is even with the surface of the rod cap.

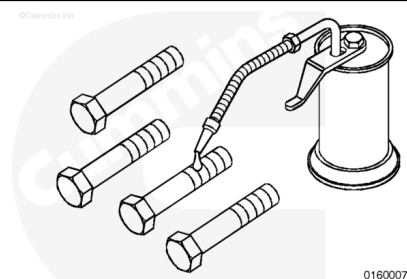
Lubricate the bearing shell with Lubriplate™ 105, Part Number 3163086.



Lubricate the connecting rod capscrew head flanges with clean gear oil Society of Automotive Engineers (SAE) W140).

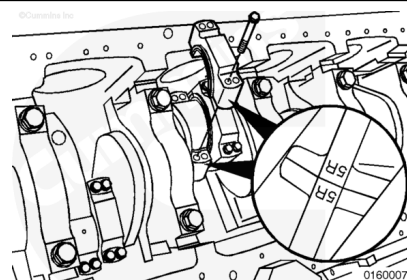
Lubricate the capscrew threads with clean engine oil (SAE 15W-40).

Install the capscrews in the rod caps.



⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The rods and the caps are machined as an assembly. The cylinder number on the rod and cap must be the same. Failure will result if the parts are mixed.



⚠ CAUTION ⚠

The side of the connecting rod cap with the cylinder number marking (bearing tang side) must be inboard on the engine. Failure will result if the parts are not installed correctly.

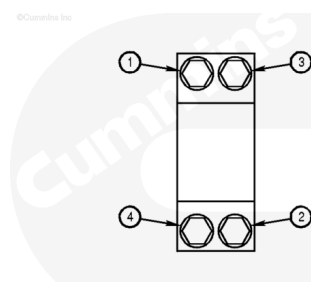
Install the connecting rod cap.

Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps and sequence to tighten the capscrews.

Use the sequence of 1, 2, 3, 4, on each connecting rod when tightening the connecting rod capscrews, as illustrated.

Use the following procedures for the connecting rod capscrew tightening specifications:

- Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-054.html)
- Refer to Procedure 001-045 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html)

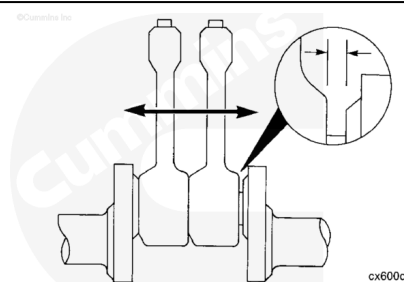


Check the side clearance between the rod and the crankshaft. Move the rods as far as they will go toward the crankshaft web. Measure the side clearance with a feeler gauge.

Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-054.html)

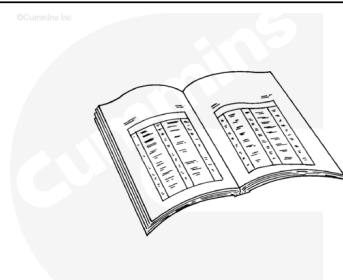
Both rods **must** move freely from side-to-side.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the lubricating oil pump, if removed. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Install the lubricating oil transfer and suction tubes and brackets. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)
- Install the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Install new oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Fill the engine with clean engine oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- On engines **not** using the PRELUB™ system, the engine **must** be manually prelubricated before operation. Refer to Procedure 007-063 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html)
- Operate the engine until the coolant temperature reaches 70°C [158°F].
- Check for coolant and oil leaks.